Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2015 JC2 Preliminary Examinations

H2 BIOLOGY **9648/2**

Paper 2: Core

Wednesday 2nd September 2015 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)
Answer **all eight** questions.
Write your answers in the spaces provided on the question paper.

Section B (Essay Question)
Answer **one** essay question only.
Write your answers on the separate answer paper provided.
All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,
1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/8
3	/7
4	/11
5	/11
6	/10
7	/12
8	/11
Total	**/80**

Section A

Answer **all** *questions*.

1. Brown adipose tissue (BAT), one of two types of fat tissue found in mammals, is highly specialized for non-shivering thermogenesis. Fig 1.1 shows how mitochondria in BAT cells function differently from those in other cells during periods of cold environmental conditions.

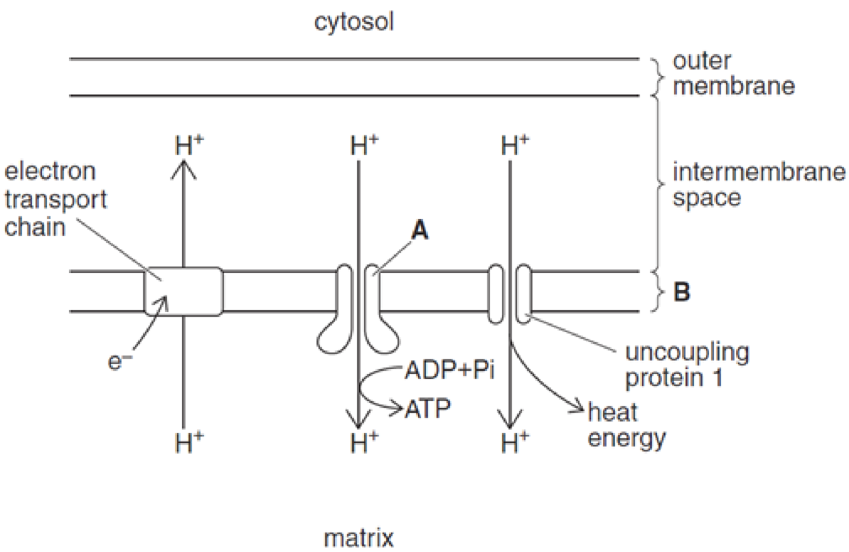

Fig. 1.1

(a) Name structures **A** and **B**.

A ……………………………

B …………………………… [2]

(b) Describe where the electrons that are passed along the electron transport chain come from.

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(c) Suggest how the uncoupling protein 1 found in the mitochondria of BAT cells is involved in thermogenesis.

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(d) Cyanide inhibits cytochrome c oxidase, which is a large transmembrane protein complex found in the electron transport chain (ETC) of mitochondria.

(i) Explain how this stops aerobic respiration.

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(ii) Explain how BAT cells can still generate ATP in the presence of cyanide.

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[Q1 Total: 10]

2. (a) *Hibiscus tiliaceus* is a common coastal plant found on the wild shores of Singapore. Fig. 2.1 shows how the amount of DNA per cell of *Hibiscus tiliaceus* varies during periods **A** to **E** in the cell cycle.

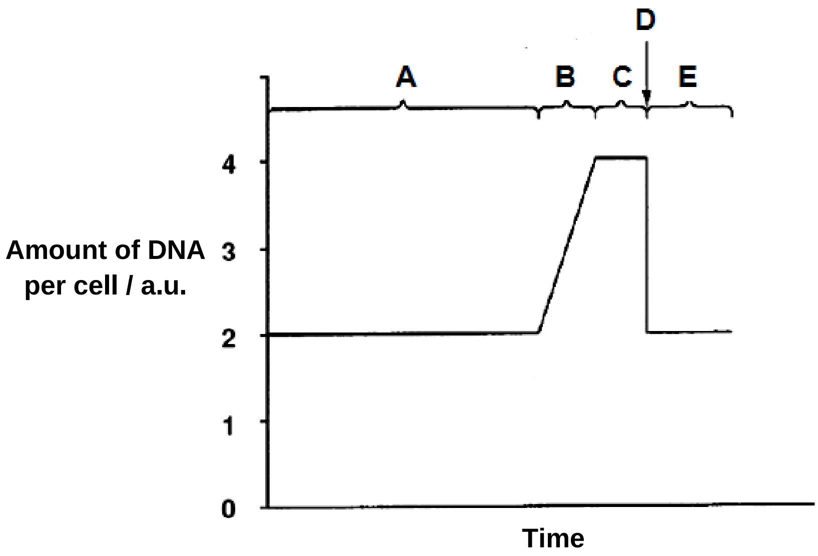

Fig. 2.1

Radioactive thymine was supplied to the cells of some growing *Hibiscus* tissue.

(i) Identify the period (**A** to **E**) during which anaphase occurs. Explain your answer.

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Fig. 2.2 shows representations of the stages of meiosis, **A** to **J** in *Hibiscus tiliaceus*, but they are not in the correct order.

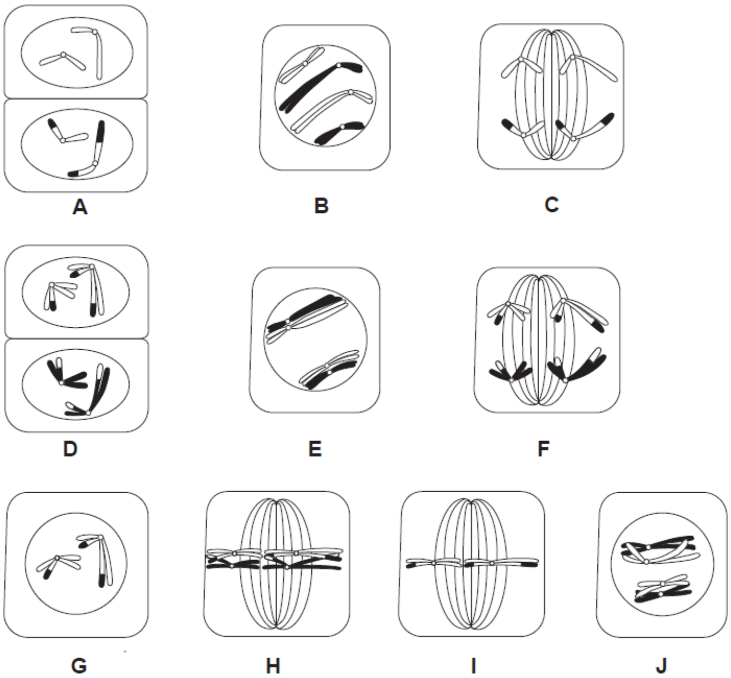

Fig 2.2

(ii) Complete the table below by writing the stages of meiosis in the correct order. Some of the stages have already been written in the table.

Meiosis **I**

B					

[1]

Meiosis **II**

G			

[1]

(iii) Explain how meiosis can result in genetic variation amongst offspring of *Hibiscus tiliaceus*.

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[Q2 Total: 8]

3.(a) In an investigation of DNA replication, a species of bacterium was grown for many generations in a medium containing ^{15}N isotope of nitrogen.

The culture was then switched to a medium containing only ^{14}N for one generation of growth; it was then returned to a ^{15}N-containing medium for two more generations of growth. DNA was then isolated from the bacterial cells, and the density of the DNA was measured using density gradient centrifugation.

In this technique, samples of different densities settle at different levels in the tube. Fig. 3.1 shows the positions of ^{15}N- and ^{14}N- labeled DNA respectively.

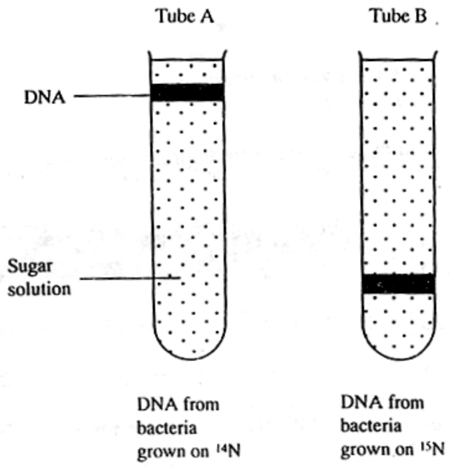

Fig. 3.1

(i) State which part of the nucleotide contains nitrogen.

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(ii) In Fig. 3.2, draw and label the pattern of the possible band(s) that can be obtained at the end of the two more generations of growth in the ^{15}**N**-containing medium. [2]

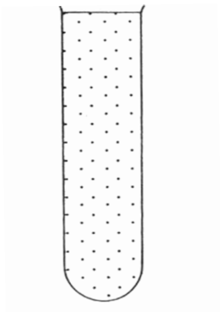

Fig 3.2

(b) The ribosomal RNA genes in human cells are located near the tips of five different chromosomes (13, 14, 15, 21 and 22). Ten small nucleoli form after mitosis in a human cell and then quickly grow and fuse to form a single large nucleolus.

Fig. 3.3 shows diagrammatically the relationship between the DNA of the ribosomal RNA genes and the nucleolus.

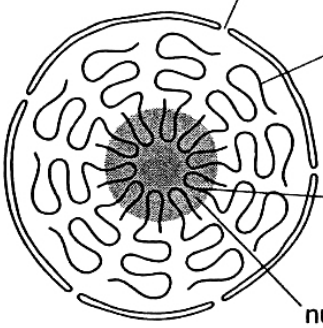

Fig. 3.3

(i) With reference to Fig. 3.3, explain why ten small nucleoli form.

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(ii) State the function of the nuclear pores in this context.

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(iii) Suggest why the nucleolus is not present during mitosis.

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[Q3 Total: 7]

4. Fig 4.1 shows a T4 bacteriophage virus infecting a bacterium.

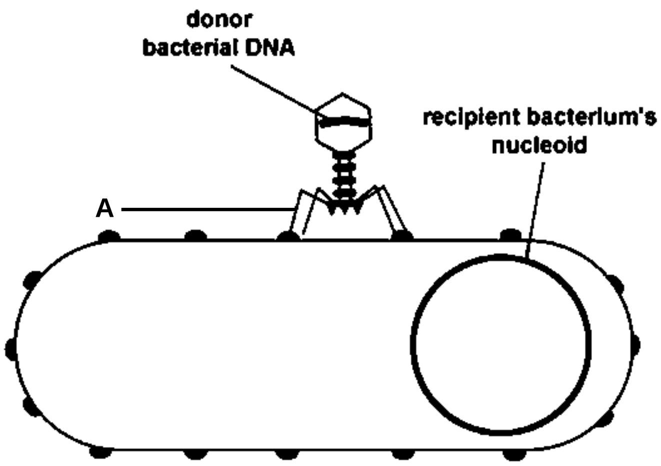

Fig. 4.1

Comment [M1]: To adjust picture contrast in greyscale to ensure printing is ok.

(a)(i) Explain how **structure A** facilitates the bacteriophage's infection process in Fig. 4.1.

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(ii) Describe how the bacteriophage in Fig. 4.1 facilitates the transfer of antibiotic resistance genes in a bacterial population.

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Fig. 4.2 shows the arabinose operon in *E.coli*. The arabinose operon has three structural genes, *araB*, *araA* and *araD*, which encode for enzymes needed for the catabolism of arabinose, a five carbon sugar that can be used by *E.coli* as an alternative carbon source. The operon has both positive and negative regulation, being activated in the presence of arabinose.

There are two operator sites (O_1 & O_2), the promoter site (situated in *araI*), and the *araC* gene which encodes for the *AraC* regulatory protein. The CAP protein functions the same way as in the *lac* operon.

(a) The arabinose operon with its regulatory gene

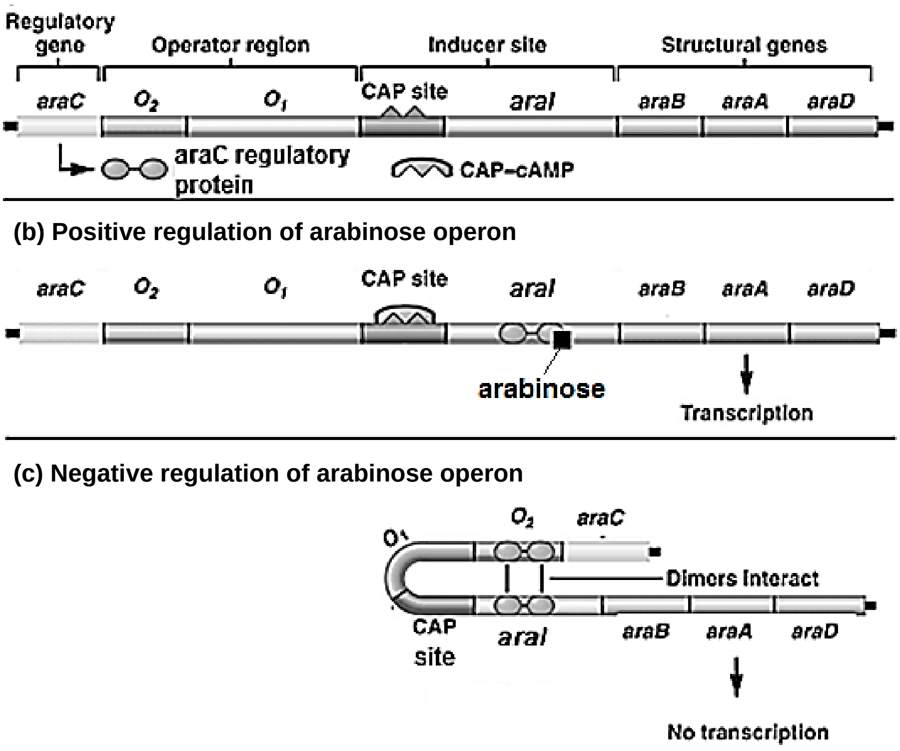

Fig. 4.2

With reference to Fig. 4.2,

(b)(i) describe and explain the **positive** regulation of the *ara* operon.

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ii) Contrast the **negative** regulation of the *ara* operon with that of the *lac* operon.

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(iii) A deletion occurred at the beginning of the ***araA*** gene. Predict the effect of this deletion on the gene product of the downstream structural gene, ***araD****.* Explain.

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[Q4 Total: 11]

5.(a) Paramyotonia congenita is a genetic disease that involves muscle stiffness and weakness. One of the causes of the disease is a single base substitution in the SCN4A gene.

The SCN4A gene codes for the alpha subunit of Na^+ channels that are abundant in skeletal muscles. The mutant SCN4A protein is full-length and permits Na^+ entry like a normal protein channel would, but the protein gate is unable to be inactivated as quickly as a normal Na^+ channel.

Suggest possible reasons for the mutated protein to remain full-length and retain the same basic function as the normal protein.

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(b) Double-stranded DNA of the SCN4A gene is denatured and allowed to hybridise through complementary base pairing with mature SCN4A mRNA isolated from skeletal muscle cells. This is shown in Fig. 5.1.

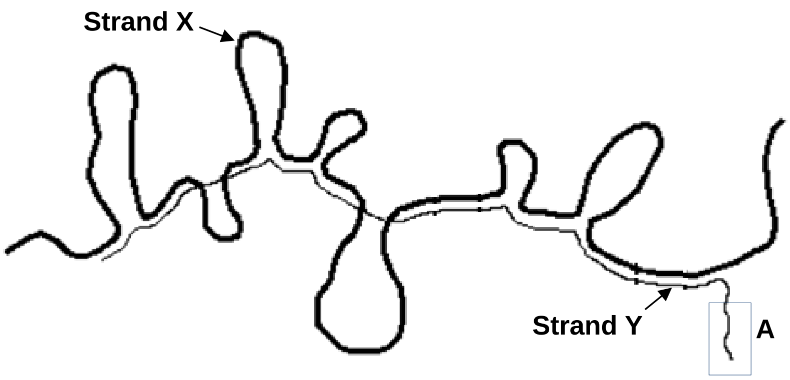

Fig 5.1

(i) State the identity of Strand **X**.

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(ii) Explain your answer in **(i)** using evidence from Fig. 5.1.

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(iii) Circle the locations of the promoter and termination regions of the SCN4A gene on Fig. 5.1. Label the regions accordingly. [2]

(iv) What does the unpaired segment (labelled **A**) on strand **Y** represent? Explain your answer.

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[Q5 Total: 11]

6. In a species of flea beetles, Phyllotreta nemorum, some individuals are parasitized by the *Hexamermis* species (a parasitic flatworm) while others have alleles that confer resistance to the parasite. Some flea beetles have also inherited the allele which codes for cellobiosidase, an enzyme that allows the individuals to feed on the toxic Winter Cress plants.

In a genetic experiment, pure-breeding flea beetles which are resistant to *Hexamermis* and are able to produce cellobiosidase were crossed with pure breeding flea beetles that are sensitive to *Hexamermis* and unable to produce cellobiosidase to produce only offspring with the ability to resist *Hexamermis* and produce cellobiosidase. When these resultant flea beetles were sibling-mated, they produced the following F_2 generation:

Resistant to *Hexamermis*, able to produce cellobiosidase	178
Resistant to *Hexamermis*, unable to produce cellobiosidase	45
Sensitive to *Hexamermis*, able to produce cellobiosidase	53
Sensitive to *Hexamermis*, unable to produce cellobiosidase	156

(a) Calculate the recombination frequency obtained from the genetic experiment.

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………[1]

(b)(i) Using the letters **A/a** (for resistance to *Hexamermis*) and **B/b** (for cellobiosidase), draw a genetic diagram to show how the F_2 generation is produced in the numbers shown above.

………[5]

Table 6.1 shows a chi-square table.

degrees of freedom	probability, p				
	0.10	*0.05*	*0.02*	*0.01*	*0.001*
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(ii) Calculate the χ^2 value.

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(iii) Using your answer in b(ii) and Table 6.1, state what conclusion may be drawn from the result.

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[Q6 Total: 10]

7(a)(i) Explain the significance of the mitochondria on the maintenance of resting potential (-70 mV) in neurones.

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Fig. 7.1 shows the electron micrograph of a junction between two neurones where neurotransmitter dopamine are released.

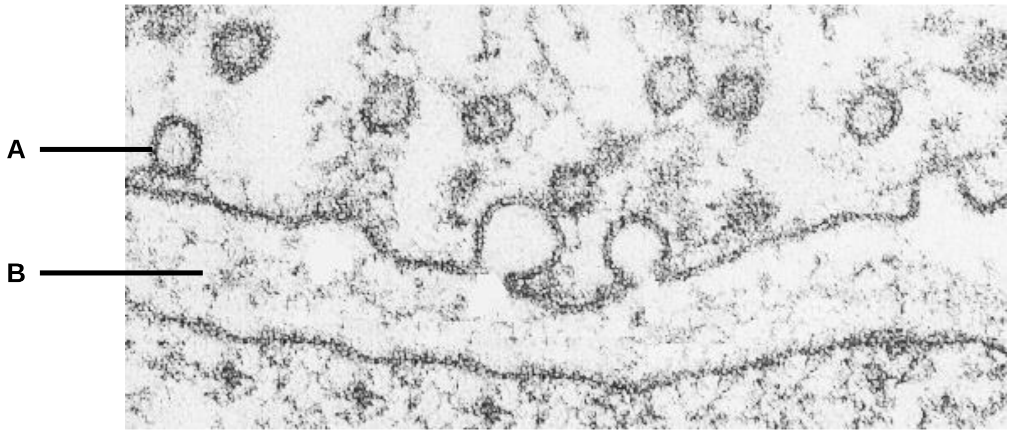

Fig. 7.1

(ii) Identify the structures labelled A and B. [1]

A: ………………………….

B: ………………………….

Abnormally high dopaminergic synaptic transmission has been linked to psychosis. Anti-psychotic medications are available to block the effects of dopamine.

(iii) Suggest two possible mechanisms of such medications.

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Heart rate is controlled by the two branches of the autonomic (involuntary) nervous system: the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). The PNS releases the hormone acetylcholine to slow the heart rate.

M2 muscarinic receptors (a part of the PNS) belong to the GPCR family. They open up K^+ channels in the heart, allowing for K^+ efflux and contributing to slowing down of heart rate. K^+ channels consist of GIRK1 and GIRK4 subunits.

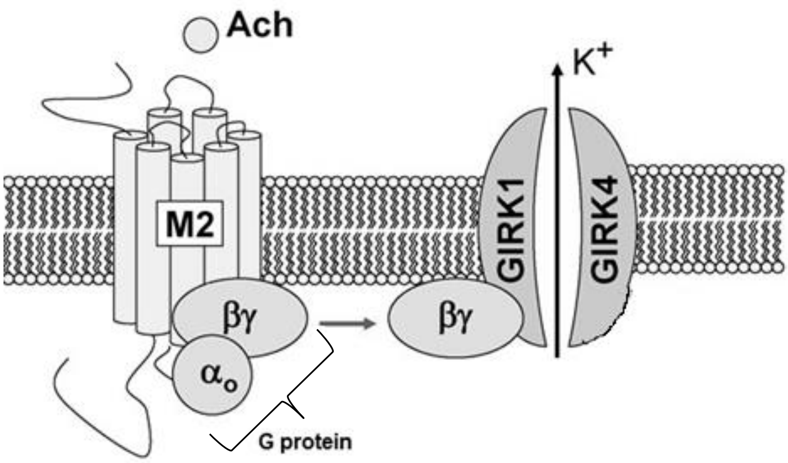

Fig. 7.2

(b)(i) Explain how the structure of the M2 muscarinic receptor allows it to be held in the membrane.

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(ii) Describe how acetylcholine slows down the heart rate.

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(c) Blood glucose level is tightly regulated in the body. Other than the changes in enzymatic activities of glycogen synthetase and glycogen phosphorylase, state two other changes that will occur within cells possessing insulin receptors when the insulin binds to the receptors.

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[Q7 Total: 12]

8. *Astyanax mexicanus* is a single species of tetra fish consisting of a sighted surface-dwelling form (surfacefish) and many blind cave-dwelling forms (cavefish) from different caves. The cavefish have small eyes that do not work, but are able to navigate in the complete darkness of the caves by emitting pressure waves with their mouths. It was found that the surfacefish were more heavily parasitized by a bacteria which infected the eyes as they do not have a thick layer of scale-like tissue covering their eyes, which is present in the cavefish.

It was thought that the ancestral morph of *Astyanax mexicanus,* which lived some 1.2 million years ago, had functional eyes. They were slowly displaced from their river habitats by large predatory fish into smaller streams that flowed through caves.

(a) Explain how natural selection could have resulted in the evolutionary change demonstrated here.

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(b) The evolution of the *Astyanax mexicanus* was demonstrated by anatomical homology. Using a named example, discuss another type of homology evidence supporting natural selection.

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(c) A fossilised *Astyanax mexicanus* specimen that dated back to 0.8 million years ago was found in the subsidiary streams of the Rio Grande. Its DNA was extracted, sequenced and compared to those obtained from the surfacefish and cavefish. It was this evidence that led scientists to classify the three members as a single species.

Explain two advantages of using molecular methods in classifying organisms.

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………[2]

[Q8 Total: 11]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

9 **(a)** Describe the molecular structure of glycogen and relate it structure to its function in living organisms. [6]

(b) Outline of the roles of cholesterol, glycolipids and glycoproteins in the cell surface membrane. [6]

(c) Describe how one bacterial cell can give rise to genetically identical cells in a colony. [8]

[Total: 20]

OR

10 **(a)** Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [8]

(b) Explain how biogeography supports the evolutionary deductions based on homologies. [6]

(c) Explain the conditions which are necessary for speciation to occur. [6]